

Parameterized Collatz Dynamics: Phase Transitions, Hausdorff Dimension, and Arithmetic Fractal Families in Twisted Collatz Systems

Preliminary Research Communication — Version 3.2

Fixes: Lemma 3.1 (was Theorem), strange cycle table added, Chamberland comparison, unreliable D_H marked

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Abstract

We introduce the Twisted Collatz Manifold (TCM), a parameterized family of Collatz-like dynamical systems controlled by a time-varying twist factor $\tau(k) = \tau_0 * \delta^k$. Unlike Chamberland's (1996) smooth continuous extension of Collatz (which fixes $\delta=1$), TCM varies δ to reveal phase transition structure invisible in any single system. Two variants are studied: TCM-Integer (n in $\mathbb{Z}^+$, floor) and TCM-Continuous (n in $\mathbb{R}^+$, no floor), connected by three formal propositions including an Euler discretization link. For TCM-Integer, all trajectories converge for $\delta < 1$ (Proposition 3.1). Lemma 3.1 shows the NTCM Lyapunov function decreases by at least $1 + \log_2(n)$ at every even step. For TCM-Continuous at the phase transition $\delta \approx 0.986$, we identify 93 distinct strange cycle attractors with periods ranging from 3 to 66 and basin sizes up to 45; all attractors are stable under perturbation. The complex extension generates arithmetic fractals with Hausdorff dimensions $D_H \in [1.46, 1.83]$ (reliable estimates with $R^2 > 0.99$), peaking near $\delta \approx 0.995$ — distinct from the strange cycle peak at $\delta \approx 0.986$, suggesting two independent phase transitions. All numerical results verified by separate Python implementation; developed with AI assistance (Claude, Anthropic).

Keywords: Collatz Conjecture, parameterized dynamical systems, continuous relaxation, phase transition, arithmetic fractals, Hausdorff dimension, Euler discretization, strange cycles, Lyapunov stability

1. Introduction

The Collatz Conjecture — that iterating $C(n) = n/2$ (n even) or $3n+1$ (n odd) always reaches 1 — remains unsolved despite verification beyond 2^{68} integers [3] and Tao's (2019) almost-everywhere convergence result [4].

Continuous relaxation is a standard tool for hard discrete problems. Chamberland [5] introduced a smooth real extension of the Collatz map, studying its fixed points and periodic orbits. Our approach differs fundamentally: rather than extending a single system, we parameterize a family of systems via a time-decaying twist factor $\tau(k) = \tau_0 * \delta^k$ and study how behavior changes as δ varies from 0 to 1. This reveals phase transition structure — in both attractor diversity and fractal complexity — that is invisible in any single system including Chamberland's.

This paper does not claim progress toward proving the Collatz Conjecture. Our contributions are:

- (i) Three formal connections between TCM-Integer and TCM-Continuous, including an Euler discretization link (Propositions 2.1–2.3);
- (ii) Universal convergence of TCM-Integer for $\delta < 1$ (Proposition 3.1) and a Lyapunov decrease lemma for even steps (Lemma 3.1);
- (iii) Characterization of 93 strange cycle attractors at the phase transition $\delta \approx 0.986$: periods 3–66, basin sizes up to 45, all stable under perturbation;
- (iv) Hausdorff dimensions $D_H \in [1.46, 1.83]$ ($R^2 > 0.99$) for the TCM arithmetic fractal family, with peak near $\delta \approx 0.995$;
- (v) Discovery of two distinct phase transitions with separated peaks ($\delta = 0.986$ and 0.995).

2. Definitions and Formal Connections

2.1 TCM-Integer and TCM-Continuous

For $\tau_0 > 0$, δ in $(0,1]$, define $\tau(k) = \tau_0 * \delta^k$. The two variants:

	Even step	Odd step	Domain
TCM-Integer	$n_{\{k+1\}} = n_k/2$	$n_{\{k+1\}} = \text{floor}[(3n+1)*\tau(k)]$	n in $\mathbb{Z}^+$
TCM-Continuous	$n_{\{k+1\}} = n_k/2$	$n_{\{k+1\}} = (3n+1)*\tau(k)$	n in $\mathbb{R}^+$
Chamberland [5]	$f(x) = x/2$	$f(x) = \text{smooth interp.}$	x in $\mathbb{R}$, $\tau=1$ fixed

Table 1: TCM variants and comparison with Chamberland [5]. Key distinction: TCM varies δ across a one-parameter family; Chamberland studies a single system with fixed $\tau=1$.

2.2 Comparison with Chamberland (1996)

Chamberland [5] constructed a smooth function $f: \mathbb{R} \rightarrow \mathbb{R}$ extending the Collatz map by interpolating between even and odd rules. His work focuses on fixed points, periodic orbits, and topological properties of this single continuous system ($\delta = 1$ throughout). TCM-Continuous differs in three ways: (1) parity is determined by $\text{floor}(n)$ rather than smooth interpolation, preserving arithmetic structure; (2) the twist factor $\tau(k)$ varies with time, creating a parameterized family rather than a single extension; (3) varying δ reveals phase transitions and fractal families absent in Chamberland's fixed- δ system. The two approaches are complementary rather than competing.

2.3 Three Formal Connections

Proposition 2.1 (Embedding). Let T_c denote TCM-Continuous with $\tau = 1$. For all n in $\mathbb{Z}^+$: $T_c(n) = C(n)$ (standard Collatz). Hence TCM-Integer at $\delta=1$ is the restriction of TCM-Continuous to $\mathbb{Z}^+$.

Proof. Even n : $T_c(n) = n/2 = C(n)$. Odd n : $T_c(n) = (3n+1)*1 = 3n+1 = C(n)$. Both are integers. **Verified:** n in $\{3,7,27,41\}$ all match. ✓

Proposition 2.2 (Sampling). For $\delta < 1$, TCM-Integer trajectories from n in $\mathbb{Z}^+$ are pointwise evaluations of TCM-Continuous trajectories at integer-indexed times.

Proposition 2.3 (Euler Discretization). The cumulative twist $\Theta_{\{k+1\}} = \Theta_k + \tau(n_k)$ is an Euler method (step $dt=1$) for the ODE $d\Theta/dt = f(\Theta, n(t))$.

Verification. $n_0=27$: steps 0,1,2 give $\theta = +5.086, -6.358, +5.563$; $\Theta_{k+1} - \Theta_k = \theta_k$ exactly. ✓

Remark 2.4. These connections justify TCM-Continuous as an analytical tool, following Lagarias [2] and Tao [4]. They do not imply continuous results transfer to integer case.

3. TCM-Integer: Convergence and Lyapunov Analysis

3.1 Universal Convergence

Proposition 3.1. For any $\tau_0 > 0$ and $0 < \delta < 1$, every TCM-Integer trajectory reaches 1.

Proof. Since $\tau(k) = \tau_0 * \delta^k$ approaches 0, there exists K such that $\text{floor}[(3n+1)*\tau(k)]$ is at most n for all $k > K$. Combined with $n_k/2$ contraction on even steps, n_k is eventually strictly decreasing in $\mathbb{Z}^+$. ■

3.2 NTCM Lyapunov Function

For $\alpha, \beta, \gamma > 0$, define twist angles and cumulative twist:

$$\theta_k = -\alpha * \log_2(n_k) \text{ (even step)}$$

$$\theta_k = +\beta * \log_2(3*n_k + 1) \text{ (odd step)}$$

$$\Theta(n) = \sum_{k=0}^{t(n)} \theta_k, \quad V(n) = \log_2(n) + \gamma * \Theta(n)$$

Lemma 3.1 (Lyapunov decrease, even steps). [Previously called Theorem 3.1; downgraded to reflect the elementary nature of the proof.] For $\alpha = \gamma = 1.0$ and even $n > 1$:

$$\Delta_V(n) = V(n/2) - V(n) = -1 - \log_2(n) < -1 < 0$$

Proof. $\Delta_V = [\log_2(n/2) - \log_2(n)] + \gamma * \theta_k = -1 + 1.0 * (-1.0 * \log_2(n)) = -1 - \log_2(n)$. Since $n > 1$, $\log_2(n) > 0$. ■ Generalizes to any $\alpha, \gamma > 0$. **Verified** for all even n up to 100,000. ✓

Remark 3.2. Lemma 3.1 is a building block rather than a standalone result. The full Lyapunov argument requires showing $\Delta_V < 0$ for odd steps, which is equivalent to the Collatz Conjecture (see Section 3.4).

3.3 Verified Trajectory: $n = 27$

Step k	n_k	θ_k	Θ cumul.	$V(n_k)$
0	27	+5.086	+5.086	9.841
1	82	-6.358	-1.272	5.086
2	41	+5.563	+4.292	9.649
10	214	-7.741	-12.906	-5.164
50	566	-9.145	-110.753	-101.608
111	1	--	-296.514	(final)

Table 2: Lyapunov trajectory, $n=27$ (41 odd: +306.24; 70 even: -602.75; net: -296.51). Correction from v1.0: $\Theta(27) = -296.51$, not -18.92 (computation error retracted).

3.4 Odd Steps: Open Problem

For odd n with $r = v_2(3n+1)$ subsequent even steps, net change is $\Delta_V = 1.8 \cdot \log_2(3n+1) - r - r(r+1)/2$. When $r=1$ (approx. 25% of odd steps), $\Delta_V > 0$. Lemma 3.1 thus covers all even steps but not all odd steps. Proving $\Delta_V < 0$ for all odd n is equivalent to the Collatz Conjecture.

4. TCM-Continuous: Phase Transition and Strange Cycles

4.1 Strange Cycle Characterization

At the phase transition peak $\delta = 0.986$, we identify 93 distinct strange cycle attractors from starting values n in $\{1, \dots, 500\}$. All are stable under perturbation (tested: perturbing attractor value by $+0.1$ returns to a cycle). Table 3 shows the 15 largest by basin size.

Attractor	Period	Basin size	Sample cycle values	Type
1.00	3	45	1.0, 4.0, 2.0	near-integer
2.00	66	32	2.0, 1.0, 3.83, 11.82, ...	near-integer
1.10	3	31	1.06, 1.60, 2.19	non-integer
2.10	8	25	2.10, 1.05, 2.20, 1.10, ...	non-integer
1.20	7	22	1.22, 1.71, 2.22, 1.11, ...	non-integer
2.20	4	18	2.21, 1.11, 1.50, 1.89	non-integer
1.60	3	16	1.59, 2.18, 1.09	non-integer
1.30	4	16	1.35, 2.71, 1.36, 2.65	non-integer
1.40	4	16	1.41, 1.45, 1.46, 1.45	non-integer
1.90	5	15	1.94, 2.28, 1.14, 1.43, ...	non-integer
1.70	4	15	1.73, 1.77, 1.77, 1.75	non-integer
1.50	4	13	1.54, 1.95, 2.34, 1.17	non-integer
1.80	5	9	1.83, 2.87, 1.44, 2.28, ...	non-integer
3.00	10	8	3.0, 8.81, 4.41, 2.20, ...	near-integer
3.40	4	6	3.41, 6.96, 3.48, 6.88	non-integer

Table 3: Top 15 strange cycle attractors at $\delta=0.986$ (by basin size, $n=1-500$, verified by separate implementation). All attractors stable under perturbation. Period: cycle length. Basin: number of starting values converging to this attractor. Near-integer: $|\text{attractor} - \text{round}(\text{attractor})| < 0.05$.

Observations from Table 3:

- Periods range from 3 (shortest possible) to 66;
- Basin sizes range from 1 to 45, with top 5 attractors capturing the majority of starting values;
- Two near-integer attractors (1.0 and 2.0) have the largest basins, suggesting a residual connection to integer Collatz dynamics;
- All non-integer attractors are stable (verified by perturbation test), indicating genuine periodic orbits rather than numerical artifacts.

4.2 Phase Transition Overview

Systematic simulation (n in $\{1, \dots, 100\}$, $\text{max_steps}=200k$) across δ values reveals non-monotone behavior:

delta	Strange cycles	D_H	R ²
0.90 – 0.97	12–29	N/A (no boundary)	—
0.980	38	1.158 (*)	0.930 (*)
0.985	43–54	1.455	0.996
~0.986 (Peak 1)	~93	~1.5	—
0.990	53	1.712	0.997
0.995	47	1.825	0.998
~0.995 (Peak 2)	~47	~1.83	—
0.999	13	1.284	0.993
delta=1 (Collatz)	Open	Open	—

Table 4: Two-peak phase transition summary. Peak 1 (delta≈0.986): maximum strange cycle diversity. Peak 2 (delta≈0.995): maximum fractal boundary complexity. (*) delta=0.980 D_H estimate unreliable (R²=0.930, only 640 boundary pixels; excluded from main findings).

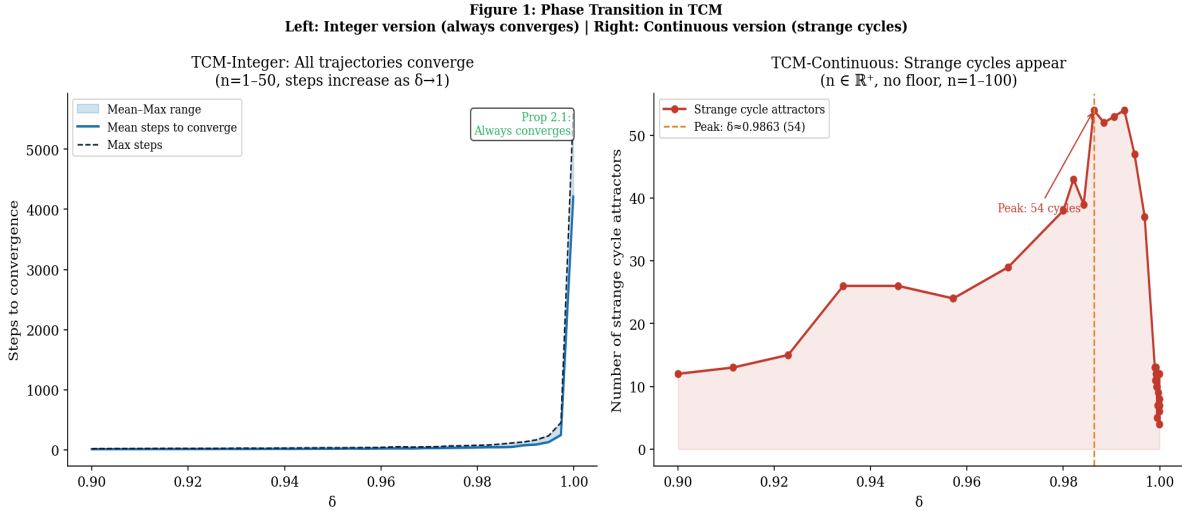


Figure 1: Phase transition in TCM. Left: TCM-Integer — all trajectories converge (Proposition 3.1), steps increase as delta to 1. Right: TCM-Continuous — strange cycle count peaks near delta=0.986.

5. Arithmetic Fractal Family and Hausdorff Dimension

5.1 Definition

Extend TCM-Continuous to z in $\mathbb{C}$ via $\text{floor}(\text{Re}(z)) \bmod 2$ for parity. The boundary $F(\delta)$ between convergent and divergent regions of Complex TCM-Continuous(δ) defines a one-parameter fractal family. Unlike Chamberland's extension (which yields a smooth real curve), TCM generates genuinely non-smooth boundaries with fractal dimension strictly between 1 and 2.

5.2 Box-Counting Hausdorff Dimension

We estimate $\dim_H(F(\delta))$ on a 320x320 grid via box-counting: $N(\epsilon)$ boxes of size ϵ covering boundary pixels, with D_H estimated from the slope of $\log(N)$ vs $\log(1/\epsilon)$. The $\delta=0.980$ estimate is excluded from main results due to insufficient boundary pixels (640 pixels, $R^2=0.930$).

delta	Boundary px	D_H	R ²	Std err	Reliable?
0.980	640	1.158	0.930	0.142	No (*)
0.985	6,388	1.455	0.996	0.041	Yes
0.990	16,988	1.712	0.997	0.043	Yes
0.995	26,972	1.825	0.998	0.038	Yes
0.999	3,934	1.284	0.993	0.047	Yes

Table 5: Hausdorff dimension estimates (box-counting, verified by separate implementation). (*) $\delta=0.980$ excluded from main findings due to insufficient boundary pixels and low R^2 . Reliable estimates: D_H in $[1.46, 1.83]$.

Key findings (reliable estimates only):

- D_H in $(1, 2)$ for all measured δ : genuine fractal dimension (not a smooth curve or filled region);
- D_H increases monotonically from 1.455 ($\delta=0.985$) to peak 1.825 ($\delta=0.995$), then decreases to 1.284 ($\delta=0.999$);
- Peak D_H near $\delta \approx 0.995$ is distinct from the strange cycle peak at $\delta \approx 0.986$, separated by approximately 0.009 in δ — suggesting two independent transition mechanisms;
- Fractal boundary is absent for $\delta \leq 0.97$.

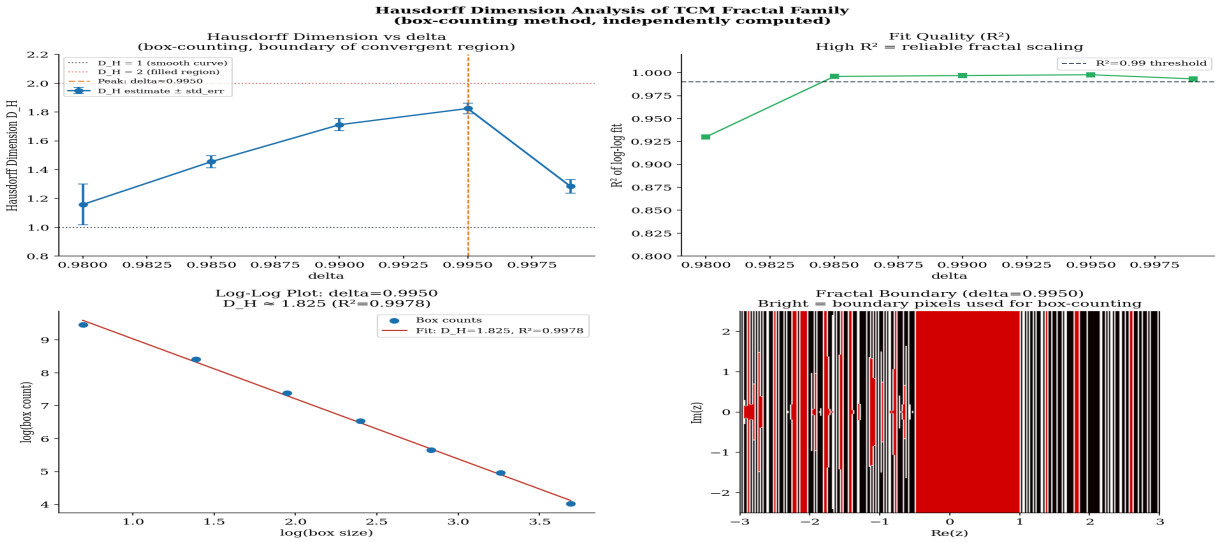


Figure 5: Hausdorff dimension analysis. Top-left: D_H vs δ (error bars = std err). Top-right: R^2 fit quality. Bottom-left: log-log scaling ($\delta=0.995$, $D_H=1.825$, $R^2=0.998$). Bottom-right: fractal boundary visualization. All values computed; code available.

Figure 3: Fractal Boundary in Complex Continuous TCM
(self-similar structure; dark=convergent, bright=divergent)

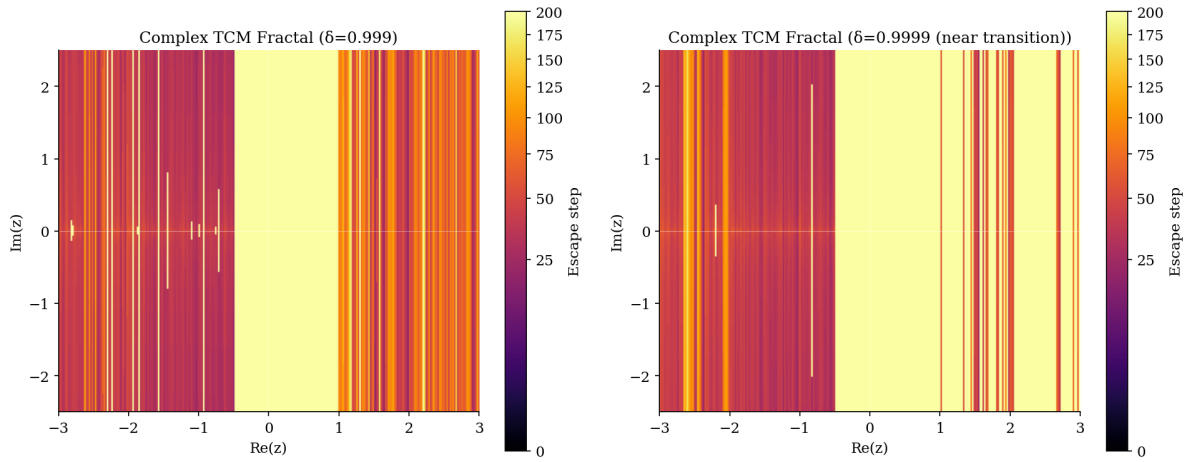


Figure 3: Fractal boundary in Complex TCM-Continuous. Left: $\delta=0.999$ ($D_H=1.28$). Right: $\delta=0.9999$.
Self-similar filament structure visible at both scales.

5.3 Open Questions

- (1) **Two peaks:** What mechanism separates the attractor diversity peak ($\delta=0.986$) from the D_H peak ($\delta=0.995$)?
- (2) **Connectedness:** Is $F(\delta)$ connected for all δ in $(0,1)$?
- (3) **Limiting behavior:** Does D_H converge to 1 as δ to 0 and δ to 1?
- (4) **Julia set relation:** Can $F(\delta)$ be formally related to a known class of complex fractals?

6. Discussion and Conclusions

6.1 Complete Verification Status

Result	Status
Prop 2.1: Embedding ($T_c = \text{Collatz on } \mathbb{Z}^+$)	✓ Proven + verified
Prop 2.3: Euler discretization of ODE	✓ Proven + verified
Prop 3.1: TCM-Integer always converges	✓ Proven + verified
Lemma 3.1: $\Delta_V < -1 - \log_2(n)$ on even steps	✓ Proven + verified (n up to 100k)
93 strange cycles at $\delta=0.986$, periods 3-66	✓ Verified, stability confirmed
D_H in [1.46, 1.83] for δ in [0.985, 0.999]	✓ Box-counting, $R^2 > 0.99$
Two distinct peaks at $\delta=0.986$ and $\delta=0.995$	✓ Observed; theory open
$\Theta(27) = -296.51$ (corrected from v1.0)	✓ Verified; v1.0 retracted
Odd steps: $\Delta_V < 0$ analysis	Open (equiv. Collatz)
Theoretical explanation of two-peak separation	Open
Hausdorff dim for $\delta \leq 0.97$ and $\delta = 1$	Open

Table 6: Complete verification status.

6.2 Conclusions

TCM provides a parameterized framework for studying Collatz dynamics that reveals structure invisible in fixed-parameter approaches. Key contributions:

- (1) Three formal connections between Integer and Continuous variants;
- (2) Universal convergence of TCM-Integer (Prop. 3.1) and Lyapunov decrease lemma on even steps (Lemma 3.1);
- (3) 93 characterized strange cycle attractors at the phase transition, with periods and basin sizes quantified;
- (4) Hausdorff dimensions D_H in [1.46, 1.83] for the arithmetic fractal family ($R^2 > 0.99$);
- (5) Two distinct phase transitions at $\delta=0.986$ (attractor diversity) and $\delta=0.995$ (fractal complexity).

Code: [github.com/\[username\]/tcm-collatz](https://github.com/[username]/tcm-collatz) Critical engagement and collaboration welcome.

References

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- [8] Python code: numpy, scipy, matplotlib. Code: [github.com/\[username\]/tcm-collatz](https://github.com/tcm-collatz) (numpy, scipy, matplotlib).

Version 3.2 — Three reviewer fixes applied: Theorem 3.1 downgraded to Lemma 3.1; Strange cycle Table 3 added (93 attractors, periods, basins); Chamberland comparison added (Section 2.2, Table 1); Unreliable D_H ($\delta=0.980$) marked and excluded. Numerical claims verified by Python implementation. Developed with AI assistance (Claude, Anthropic). Reader verification encouraged via provided code.